Azure SQL Database

Licensing & Pricing — Complete Guide

Soumya Kanta Panda

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DTU Model • vCore Model • Serverless • Azure Hybrid Benefit • Scaling Strategies

1. What is Azure SQL Database?

Azure SQL Database is a fully managed PaaS (Platform as a Service) relational database engine built on the latest stable Microsoft SQL Server.

- Microsoft handles ALL infrastructure: hardware, OS patching, backups, high availability, and disaster recovery.
- You pay per database per month — focus only on your schema, queries, and application logic.
- Supports both on-premises migration and modern cloud-native application development.
- Delivers a 99.99% SLA with built-in redundancy and automatic failover.

Microsoft Manages vs. You Manage

MICROSOFT MANAGES	YOU MANAGE
✓ Hardware & Servers	✓ Database Schema
✓ SQL Server Engine	✓ Application Logic
✓ OS Patching	✓ Data Security & Encryption
✓ Automated Backups	✓ Performance Tuning
✓ High Availability	✓ Access Control & Firewall Rules
✓ Storage Management	✓ Monitoring & Alerts
✓ Security Patches	✓ Tier / Model Selection

2. DTU Model — Database Transaction Unit

A DTU is a blended metric combining CPU + Memory + Read IOPS + Write IOPS into a single number.

- Think of it like a horsepower rating for your database — more DTUs = more compute power.
- DTUs range from 5 (Basic) to 4,000 (Premium). Storage is INCLUDED in the price.
- Best for: databases up to 4 TB, dev/test environments, teams wanting predictable monthly bills.
- Limitation: You cannot see underlying CPU type, RAM, or IOPS — it is an abstracted black-box metric.
- IMPORTANT: Azure Hybrid Benefit is NOT available for the DTU model.

DTU Service Tiers

Tier	DTU Range	Best Use & Details	Storage	Monthly Cost
Basic	**5**	**Use:** Very small DBs, learning, testing, POCs **Detail:** Simplest entry point. Fixed 5 DTUs. No scale slider.	Up to 2 GB	**~$4.90 / month**
Standard	**10 – 3,000**	**Use:** Medium apps, dev/test, batch processing **Detail:** Most popular. Scale DTUs up/down without downtime. Good cost/performance.	Up to 1 TB	**~$14.72 – $4,416 / month**
Premium	**125 – 4,000**	**Use:** Business-critical, high IOPS, low latency **Detail:** In-memory OLTP. Highest IOPS. Zone redundant. For high-traffic production.	Up to 4 TB	**~$456 – $15,699 / month**

3. vCore Model — Virtual Core

A vCore is a virtual CPU core — you choose the EXACT number of cores, and memory scales automatically alongside it.

- Unlike DTU, vCore gives you FULL transparency: actual CPU generation, RAM amount, and IOPS are visible.
- Compute and Storage are billed SEPARATELY — scale them independently for optimal cost control.
- Two compute modes: Provisioned (fixed cores, billed hourly) and Serverless (auto-scale, billed per second).
- ONLY vCore Provisioned supports Azure Hybrid Benefit — bring your SA licenses to save up to 30%+.

vCore Service Tiers

General Purpose (GP)

vCores	Memory	IOPS	Price
2 – 80 vCores	**Up to 408 GB RAM (5.1 GB/vCore)**	**Up to 20,000 IOPS**	**From ~$0.504/hr (2 vCores, East US)**

Storage: Up to 4 TB | Remote SSD

Hardware: Standard-Series (Gen5), Fsv2-Series, DC-Series

Best For:

- Standard OLTP applications & web backends
- Medium-complexity analytics workloads
- Dev / Test production-like environments
- Cost-balanced general business workloads

Hyperscale (HS)

vCores	Memory	IOPS	Price
2 – 80+ vCores	**Up to 1,024 GB RAM**	**Up to 204,800 IOPS**	**From ~$0.366/hr (2 vCores, East US)**

Storage: Up to 100 TB | Distributed page-server architecture

Hardware: Standard-Series, Premium-Series Memory-Optimized

Best For:

- Very Large Databases (VLDBs) exceeding 4 TB
- High IOPS data warehousing workloads
- Rapid horizontal scale-out via read replicas
- Near-instant backups using snapshot technology

Business Critical (BC)

vCores	Memory	IOPS	Price
2 – 128 vCores	Up to 625 GB RAM	Up to 200,000 IOPS (local SSD)	From ~$1.35/hr (2 vCores, East US)

Storage: Up to 4 TB | Local NVMe SSD (ultra-fast)

Hardware: Standard-Series (Gen5), DC-Series, M-Series

Best For:

- Mission-critical production systems
- Ultra-low latency and high-resilience needs
- Always-On availability + zone redundancy built-in
- One free readable read-scale replica included

4. Serverless Compute Tier

Serverless is a compute tier within the vCore model that automatically scales CPU between a minimum and maximum you define.

- Billing: Charged per vCore-second of ACTUAL compute used — not for idle time. Rate: $0.5218/vCore-hour.
- Auto-Pause: Database pauses after configurable inactivity delay (min 1 hr). Only storage is billed during pause.
- Auto-Resume: Automatically resumes on next connection — first query may see a slight cold-start latency.
- Scale range: 0.5 – 16 vCores. Memory scales with compute (2.02 GB – 48 GB). Storage billed separately.
- Note: Serverless does NOT support Azure Hybrid Benefit. Best for intermittent / unpredictable workloads.

Serverless vs Provisioned — How Billing Works

PROVISIONED COMPUTE	SERVERLESS COMPUTE
Fixed vCores — Billed Every Hour Regardless of Activity	Auto-scales — Billed per Second of Actual Use
Active period: BILLED	**Active period: BILLED (per second)**
Idle period: STILL BILLED — paying for nothing!	**Idle period: AUTO-PAUSED — FREE!**
Cost is FIXED — idle periods waste money.	Cost follows ACTUAL use — idle = near zero cost.

5. Azure Hybrid Benefit (AHB)

Azure Hybrid Benefit lets you bring existing on-premises SQL Server licenses (with Software Assurance) to Azure SQL Database.

- Instead of paying the full 'License Included' price, you pay only base compute cost — SQL license cost is waived.
- Savings: Up to 30%+ on Azure SQL Database. Combined with Reserved Instances: savings can reach 55%+.
- Eligible: SQL Server Enterprise or Standard Edition with active Software Assurance (SA).
- Applies to: vCore Provisioned compute tier ONLY. Not available for DTU or vCore Serverless.
- Enterprise Edition bonus: Every 1 on-prem core entitles you to 4 vCores of General Purpose in Azure.

How AHB Works

Step 1
On-Premises
SQL Server License
+ Software Assurance

→

Step 2
Azure Portal
Compute + Storage Settings
"Save Money" toggle ON

→

Result
Reduced Cost!
Pay: Base Compute Only
SQL License Cost = $0

AHB Alone	AHB + 1yr Reserved	AHB + 3yr Reserved
Save up to 30%+	**Save up to 45%**	**Save up to 55%+**

6. DTU vs vCore — Complete Head-to-Head Comparison

Criteria	DTU Model	vCore Model
Billing Basis	Fixed bundle: compute + storage + I/O	Separate: compute billed hourly OR per-second
Cost Transparency	Abstracted metric — black box, no visibility	Full visibility: CPU type, RAM, IOPS breakdown
Storage	INCLUDED in price (up to 4 TB max)	SEPARATE billing — up to 100 TB (Hyperscale)
Compute Scaling	Scale DTU count up / down (bundled)	Scale vCores & storage independently
Azure Hybrid Benefit	**NOT SUPPORTED (no license savings)**	**SUPPORTED on Provisioned tier (save 30%+)**
Serverless Option	**NOT AVAILABLE**	**AVAILABLE (auto-scale, auto-pause)**
Max Compute	4,000 DTUs	128 vCores (Business Critical)
Max Storage	4 TB (Premium tier)	100 TB (Hyperscale tier)
In-Memory OLTP	Premium tier only	Business Critical tier
Read Replicas	Geo-replication (billed separately)	1 free replica (Business Critical)
Zone Redundancy	Premium tier (extra cost)	All vCore tiers supported
Best For	Dev/Test, small DBs, fixed billing, no SA	Production, VLDBs, cost-control, SA holders

7. When to Scale Your Azure SQL Database — 4 Key Signals

Signal 1 CPU / DTU Usage > 70%	Signal 2 Storage Usage > 80%	Signal 3 Large Data Load Incoming	Signal 4 Query Latency / Timeouts
• Sustained compute above 70% means your database is near saturation. • Even short bursts above 90% cause query timeouts and slowdowns. • Action: Scale up BEFORE usage hits 100% — do not wait.	• When data exceeds 80% of provisioned storage, throttling can occur. • A storage-full state causes write failures and application errors. • Action: Scale storage (vCore) or move to a higher DTU tier immediately.	• ETL jobs, data engineering pipelines, and bulk imports need more compute. • Seasonal spikes (quarter-end, holiday traffic) are predictable — plan ahead. • Action: Pre-scale during off-peak hours BEFORE the load arrives.	• Queries that took seconds now take minutes — a clear saturation warning. • Connection pool exhaustion and timeout errors appear in application logs. • Action: Identify top resource-consuming queries AND scale up together.

8. How to Monitor & Scale — Azure Portal Steps

How to Monitor DTU / vCore Usage

1	**Open Azure Portal**	Go to portal.azure.com and sign in with your Azure credentials.
2	**Find Your SQL Server**	Use the top search bar — type SQL Server name and select from 'SQL Servers'.
3	**Select Your Database**	Left panel → Settings → click 'Databases' → click target database.
4	**Open Monitoring Tab**	Database Overview page → scroll to 'Monitoring' → view DTU/vCore % graph.
5	**Choose Time Range**	Click the time selector: options are 1 hr / 24 hrs / 7 days.
6	**Check Storage**	Same Overview page → find 'Size consumed' and 'Space remaining'.

How to Scale Up / Scale Down

1	**Go to SQL Server**	Azure Portal → All Resources → select your SQL Server.
2	**Compute + Storage**	Left sidebar → Settings → click 'Compute + storage'.
3	**Select Tier / Model**	Choose DTU or vCore, then select the service tier.
4	**Set Compute Size**	Drag the slider or type the DTU count / vCore count. Preview monthly cost.
5	**Set Storage (vCore)**	vCore only: set storage size in GB and choose backup redundancy.
6	**Apply Changes**	Click 'Apply' / 'Save'. Scaling is usually zero-downtime.*

** Scaling from Basic → Standard / Premium may take a few minutes. Always scale during off-peak hours.*

9. DBA Decision Framework — Which Model Should YOU Choose?

Choose DTU	Smaller databases up to 4 TB | Dev/Test environments | Fixed-cost billing preferred | No SQL Server Software Assurance
Choose vCore Provisioned	Production workloads | Need CPU/RAM transparency | Large databases | Have SQL SA licenses → activate Hybrid Benefit!
Choose vCore Serverless	Intermittent workloads | Dev environments | Want auto-pause savings | Cannot predict usage patterns
Choose Hyperscale	Databases > 4 TB | Rapid scale-out reads needed | High IOPS | Cloud-native VLDB design expected

Source: Microsoft Azure Official Documentation & Microsoft Learn | azure.microsoft.com